

Agricultural Advisory QA System

A Retrieval-Augmented Transformer for Farmer Query Answering

Abbareddy Sahasra
Ravi Abhinav
Rendla Venkata Pranav
Dayanithaa Murughanandam
Vinay Reddy

April 26, 2026

Drive Link: [Click here](#)

GitHub Repo: [Click here](#)

1 Introduction

This report presents an Agricultural Question Answering (QA) system built using the PyTorch framework. The system is trained on a combination of Q&A datasets covering agricultural topics such as crops, pests, nutrients, and irrigation, with the primary dataset being the Kisan Call Centre (KCC) corpus.

The core model is a custom sequence-to-sequence Transformer that takes a farmer's question as input and generates a relevant advisory answer. To improve the quality and factual grounding of responses, the system includes a Retrieval-Augmented Generation (RAG) pipeline that fetches relevant knowledge from the dataset at query time and provides it as additional context to the model.

1.1 Objectives

- Design and train a Transformer-based encoder-decoder model for agricultural QA.
- Initialise token embeddings from BERT to leverage pre-trained linguistic knowledge.
- Implement a RAG pipeline using the model's own encoder for dense retrieval.

2 Dataset

The primary dataset is the KCC corpus comprising over 130,000 verified agricultural Q&A pairs, supplemented with other datasets.

Column	Type	Description
--------	------	-------------

question	string	Farmer’s natural-language query
answer	string	Expert advisory answer
intent	string	Query category
state	string	Indian state
crop	string	Crop name

Rows with missing values are dropped. Dataset split: 95% train / 5% validation.

3 Tokenisation

A Byte-Pair Encoding (BPE) tokenizer is trained using HuggingFace.

Token	ID	Purpose
<PAD>	0	Padding
<SOS>	1	Start of sequence
<EOS>	2	End of sequence
<UNK>	3	Unknown token
[CTX]	4	RAG context separator
[Q]	5	Question separator

Vocabulary size: 50,000 tokens.

4 Embeddings

The model uses BERT (`bert-base-uncased`) static embeddings instead of random initialization.

5 Model Architecture

TransformerQA is an encoder–decoder Transformer with ~ 51 M parameters.

Hyperparameter	Value	Notes
vocab size	50,000	BPE vocab head dim = 64
hidden dim	384	
encoder layers	4	
decoder layers	4	
heads	6	
ff dim	1024	
dropout	0.15	

5.1 Textual Pipeline

1. Token embedding + positional encoding
2. Encoder (multi-head attention + FFN)
3. Encoder output passed to decoder

4. Decoder (masked + cross attention)
5. Output projection and beam search

6 Training

- Optimiser: AdamW
- LR schedule: Warmup-cosine
- Loss: Cross-entropy with label smoothing

Parameter	Value
Batch size	32
Epochs	15
Split	95/5
Label smoothing	0.1

6.1 RAG-Augmented Training

30% of samples include RAG context:

$$[CTX] \ c_1 \ [CTX] \ c_2 \ \dots [Q] \ q$$

7 Retrieval-Augmented Generation (RAG)

7.1 Overview

The RAG pipeline allows the model to condition its answers on a dynamically retrieved knowledge base rather than relying solely on memorised weights. This is especially important for a domain like agriculture where crop-specific advice must be precise and verifiable.

The RAG system consists of three components:

1. **Knowledge Base Builder** (`rag_knowledge_base.py`): Formats raw CSV rows into searchable text chunks.
2. **Retriever** (`rag_retriever.py`): Encodes chunks and queries using the model's own encoder and performs dense similarity search.
3. **RAG Inference** (`rag_inference.py`): Retrieves context, constructs augmented input, and generates an answer.

7.2 Retrieval Thresholds

Parameter	Value	Description
-----------	-------	-------------

rag_min_similarity	0.50	Minimum cosine similarity required to accept a retrieved chunk
rag_top_k	5	Maximum number of chunks retrieved per query
rag_max_seq_len	320	Encoder input length for RAG-augmented sequences

Chunks falling below the similarity threshold are discarded. In such cases, the model falls back to generating answers based purely on its learned parameters.

7.3 Augmented Encoder Input

Retrieved chunks are prepended to the original question using special separator tokens:

$$[CTX] c_1 [CTX] c_2 \dots [CTX] c_k [Q] q$$

This sequence is tokenised and padded to 320 tokens before being fed to the encoder. The decoder then attends over the entire encoder output via cross-attention, allowing it to utilise both retrieved context and the original question.

7.4 Training Integration (Curriculum RAG)

During training, RAG augmentation is applied to only a subset of the data:

- 30% of training samples use RAG-augmented input
- 70% use plain question input
- The subset is reshuffled every epoch

This curriculum strategy prevents over-reliance on retrieval and ensures the model can:

- Answer effectively using retrieved context
- Fall back to internal knowledge when retrieval is weak or unavailable

8 Evaluation

Metrics used:

- Token accuracy
- BLEU
- ROUGE (1, 2, L)

9 System Components

File	Responsibility
config.py	Hyperparameters
data_loader.py	Tokenizer + dataset
transformer.py	Model
train.py	Training loop
rag_retriever.py	Retrieval
rag_inference.py	RAG inference
evaluate_model.py	Metrics

10 Key Design Decisions

1. BERT embeddings improve convergence
2. Shared encoder for retrieval and generation
3. 30% RAG curriculum training
4. Warmup-cosine LR stabilises training

11 Results

The following figures show sample outputs from the chatbot system.

```
Farmer > what are common diseases in wheat?
[Debug] Raw IDs: [9478, 454, 1086, 1484]
Bot    > Viral and bacterial diseases

Farmer > what is the fertilizer schedule for wheat?
[Debug] Raw IDs: [481, 772, 479, 544, 43, 667, 479, 651, 454, 51, 44, 52, 479, 459, 558, 45]
Bot    > apply 12 kg urea , 25 kg ssp and 4 - 5 kg per acre .

Farmer > why are pulse plants not growing properly?
[Debug] Raw IDs: [1810, 428, 6091, 423, 1110, 8954, 418, 434, 689, 45]
Bot    > due to lot of sugar levels in the soil .

Farmer > how to improve cotton yield?
[Debug] Raw IDs: [1462, 5435, 692, 423, 828, 699, 2680, 1453, 1134, 5347, 43, 1679, 434, 1171, 423, 3297, 2458, 1036, 418, 434]
Bot    > take specific variety of cotton that makes your yield potential , increase the number of boll worms infestation in the

Farmer > what diseases affect chilli plants?
[Debug] Raw IDs: [9695, 3181]
Bot    > Bean rust

Farmer > why are chilli plants not flowering?
[Debug] Raw IDs: [5301, 583, 496, 1810, 428, 655, 7042]
Bot    > This can be due to nutrient factors
```

Figure 1: Chatbot output example 1

```

Agricultural QA Chatbot
=====
Initializing system... (please wait)
Loading existing BPE vocabulary from vocab_bpe.json (size=50000)
Loaded trained weights from 'transformer_qa.pth'! (155/155 tensors)

Chatbot is ready! Type 'quit' or 'exit' to stop.
-----

Farmer > best fertilizer for wheat
[Debug] Raw IDs: [481, 1359, 520, 479, 43, 640, 667, 479, 454, 1270, 2088, 46, 573]
Bot    > apply dap 50 kg , mop 25 kg and fym 10kg / bigha

Farmer > seed treatment for blackgram
[Debug] Raw IDs: [743, 48, 479, 423, 815, 549, 49, 45, 52, 102, 423, 1090, 454, 52, 465, 423, 2479, 107, 42804, 454]
Bot    > treat 1 kg of seeds with 2 . 5 g of mancozeb and 5 ml of imidach l ropid and

Farmer > what causes leaf curl in chilli
[Debug] Raw IDs: [14089, 1521, 4194, 1642]
Bot    > Insects lay eggs there

```

Figure 2: Chatbot output example 2

12 Conclusion

This system combines a Transformer model with BERT embeddings and a RAG pipeline to produce grounded agricultural answers. It balances retrieval-based and memory-based responses effectively.